

RV COLLEGE OF ENGINEERING

DEPARTMENT OF AEROSPACE ENGINEERING

Structural Mechanics

Author:
Benjamin Rohit

*A compilation of notes/questions available on structural mechanics. Only for use at
the department of Aerospace Engineering, RVCE*

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“Thanks to my solid academic training, today I can write hundreds of words on virtually any topic without possessing a shred of information, which is how I got a good job in journalism.”

Dave Barry

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Chapter 1

Basic equations of linear elasticity and Principal Stresses and Strain

1.1 Question Bank

Quiz Questions

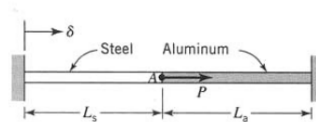
1. The deformation of a bar under its own weight as compared to that when subjected to a direct axial load equal to its own weight will be....
2. The highest stress that a material can withstand for a specified length of time without excessive deformation is called
3. The stretch in a steel rod of circular section having length L subjected to a tensile load P and tapering uniformly from diameter D_1 to D_2 is given as
4. A block of steel is loaded by a tangential force on its top surface while the bottom surface is held rigidly. The deformation of the block is due to.....
5. When tensile stress is applied axially on a circular rod its diameter.....
6. For a linearly elastic, isotropic and homogeneous material, the number of elastic constants required to relate stress and strain is.....
7. The material that exhibits the same elastic properties in all directions at a point is said to be.....
8. What is an incompressible medium, and what is its Poisson's ratio?
9. The relation between the stiffness matrix K and the Compliance Matrix C is
10. The Young's modulus of elasticity of a material is 2.5 times its modulus of rigidity. The Poisson's ratio for the material will be:.....
11. The relation between E , G , K and μ are.....
12. Write the Compliance matrix for an Anisotropic Material.
13. For any engineering material, Poisson's ratio limiting values are between.....
14. If Elastic modulus of material is 200 Gpa and Poisson's ratio is 0.29, bulk modulus is....
15. Load divided by initial cross-section area of specimen before deformation is called.....

16. If design load of a material is 20kN and factor of safety is two, then ultimate stress will be.....
17. In a general two dimensional stress system, planes of maximum shear stress are inclined at with principal planes.
18. If a piece of material neither expands nor contracts in volume when subjected to stress, then the Poissons ratio must be.....
19. The phenomenon of progressive extension of the material i.e., strain increasing with time at a constant load is called.....
20. The number of components in a stress tensor defining stress at a point in three dimensions is.....
21. Two examples of plane stress conditions are.....
22. The number of elastic constants for a completely anisotropic elastic material which follows Hooke's law is.....
23. Materials which show direction dependent properties are called.....
24. In terms of Poisson's ratio the ratio of Young's Modulus (E) to Shear Modulus (G) of elastic materials is.....
25. A weight falls on a plunger fitted in container filled with oil thereby producing a pressure of 1.5 N/mm² in the oil. The bulk modulus of oil is 2800 N/mm². Given this situation the volumetric compressive strain produced in the oil will be
26. A cube with a side of length of 1cm is heated uniformly to 1° C above the room temperature and all sides are free to expand. What will be the increase in volume of the cube? Assume thermal expansion is α per ° C
27. what is an inelastic material?
28. Two tapering bars of the same material are subjected to a tensile load of P. The lengths of both the bars are the same. The larger of each of the bars is D. The diameter of the bar A at its smaller end is D/2 and that of the bar B is D/3. What is the ratio of elongation of the bar A to that of the bar B?
29. For any engineering material, poissons ratio limiting values are between
30. If elastic modulus of a material is 200 GPa and poissons ratio is 0.29, then bulk modulus is
31. Load divided by initial cross section area of specimen before deformation is called
32. If a material is subjected to two incremental true strains namely ϵ_1 and ϵ_2 , then the total true strain is
33. The relationship between stress and strain defined in Hooke's law is independent of
34. The radius of a wire stretched by load is reduced by half the Young's modulus will be

35. What would be the Poisson's ratio of an incompressible medium?
36. If μ is 0.5 the ratio of G to E will be

Main Questions

1. Derive the Generalized Hooke's law in the Compliance form and, Write the compliance matrix.
2. Derive the relationship between the linear elastic properties Young's modulus (E), rigidity modulus (G) and bulk modulus (K)?
3. What different stresses are set-up in a bolt due to initial tightening, while used as a fastener? Name all the stresses in detail.
4. Draw the stress-strain curve for a ductile material and explain salient points on the curve.
5. A steel bar and an aluminum bar are joined end to end and fixed between two rigid walls as shown in Figure. The cross-sectional area of the steel bar is A , and that of the aluminum bar is A_1 . Initially, the two bars are stress free. Derive general expressions for the deflection of point A, the stress in the steel bar, and the stress in the aluminum bar when
 - (a) Load P is applied at point A
 - (b) The left wall is displaced an amount δ to the right.



6. Prove that shear stress τ increases to a maximum value at 45 degrees when a uni-axial stress is applied over an oblique plane.
7. What are principle planes? Explain Principle stresses and strains and what is their significance.
8. Find the principal stresses and the corresponding directions from the stress tensor.

$$\sigma = \begin{bmatrix} 3 & 4 & 6 \\ 4 & 2 & 5 \\ 6 & 5 & 1 \end{bmatrix}$$

9. Derive the expression for axial deformation of a bar with constant cross section. A steel wire 2m long and 3 mm in diameter is extended by 0.75 mm. When a weight W is suspended from the wire. If the same weight is suspended from a brass wire, 2.5 m long and 2 mm in diameter. It is elongated by 4.64 mm. Determine the modulus of elasticity of brass if that of steel be $2 \cdot 10^5 \text{ N/mm}^2$
10. A Magnesium plate in Biaxial stress is subjected to tensile of 24 Mpa in X direction, 12Mpa in the Y direction. The corresponding strains in the plate are 440×10^{-6} and 80×10^{-6} in the X and Y direction respectively. Determine the Poisson's ratio and the Modulus of Elasticity.

11. A steel bolt of 20 mm diameter passes centrally through a copper tube of internal diameter 28 mm and external diameter 40 mm. The length of whole assembly is 600 mm. After tight fitting of the assembly, the nut is over tightened by quarter of a turn. What are the stresses introduced in the bolt and tube, if pitch of nut is 2 mm? Take $E_s = 2 \cdot 10^5 \text{ N/mm}^2$ and $E_c = 1.2 \cdot 10^5 \text{ N/mm}^2$.
12. A cylindrical specimen of a titanium alloy having an elastic modulus of 108 GPa and an original diameter of 3.9 mm will experience only elastic deformation when a tensile load of 2000 N is applied. Compute the maximum length of the specimen before deformation if the maximum allowable elongation is 0.42 mm. And list at least two factors that promote transition from ductile to brittle fracture.
13. A composite rod is 1000 mm long, its two ends are 40 mm^2 and 30 mm^2 in area and length are 300 mm and 200 mm respectively. The middle portion of the rod is 20 mm^2 in area and 500 mm long. If the rod is subjected to an axial tensile load of 1000 N, find its total elongation. ($E = 200 \text{ GPa}$).
14. Derive the relationship for a deflection of a bar under uniaxial tension. The assembly consists of two 10-mm diameter red brass C83400 copper rods AB and CD, a 15-mm diameter 304 stainless steel rod EF, and a rigid bar G. If $P = 5 \text{ kN}$, determine the horizontal displacement of end F of rod EF. (Take E for brass to be 101 GPa and E for steel to be 193 GPa)
15. A bar of length L tapers uniformly from diameter $1.1 D$ at one end to $0.9 D$ at the other end. The elongation due to axial pull is computed using mean diameter D . What is the approximate error in computed elongation?
16. A cylindrical specimen of a titanium alloy having an elastic modulus of 108 GPa and an original diameter of 3.9 mm will experience only elastic deformation when a tensile load of 2000 N is applied. Compute the maximum length of the specimen before deformation if the maximum allowable elongation is 0.42 mm.
17. A metallic bar $300 \times 100 \times 40$ is subjected to a force of 5kN, 6kN and 4kN along x, y and z directions respectively. Evaluate the change in the volume of the block. Take youngs modulus to be 200GPa and poissons ratio as 0.25.
18. Explain the significance of constitutive equations in elasticity problems.
19. A bar of circular cross section having two different diameters D and $2D$, have lengths of $L/2$ each. Calculate the strain energy if load $P=27\text{kN}$, length $L=600\text{mm}$ and diameter $D=40\text{mm}$ and $E=105\text{GPa}$.
20. Prove that $\sigma_{x'x'} + \sigma_{y'y'} = \sigma_{xx} + \sigma_{yy}$
21. Prove that in a two dimensional stress system, planes of maximum shear stress are inclined at 45° with the principal planes.

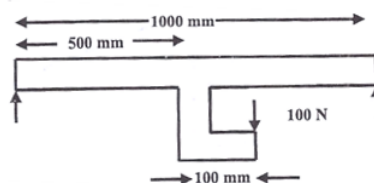
Chapter 2

Bending Moment and Shear Force Diagram

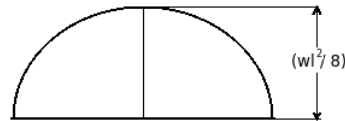
2.1 Question Bank UNIT - 2

Quiz Questions

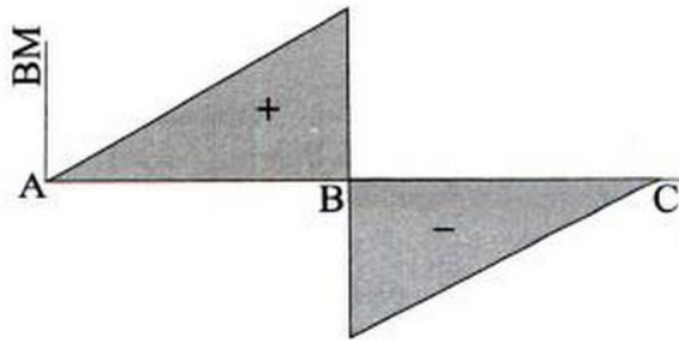
1. A simply supported beam of span length 6m and 75mm diameter carries a UDL of 1.5 kN/m. What is the maximum value of bending moment? (6.75kN-M)
2. A sudden jump anywhere on the bending moment diagram of a beam is caused by a?(*Moment*)
3. In a cantilever subjected to a combination of concentrated load, UDL, UVL, the maximum bending moment will occur at(*fixed end*)
4. In a simply supported beam carrying a uniformly distributed load over the left span indicates that the point of contra-flexure will occur at(*no point of contra-flexure*)
5. When the bending moment is parabolic curve between two points it indicates that there is a load (*UDL*)
6. What is the maximum bending moment for simply supported beam carrying a point load "W" kN at its centre?($WL/4$)
7. In a simply-supported beam loaded as shown below, the maximum bending moment is Nm is



8. Uniformly varying load between two sections in shear force diagram is represented by
9. In bending moment diagram, if no load acts between two sections, then it is represented by.....
10. The diagram shown below, depicts bending diagram for

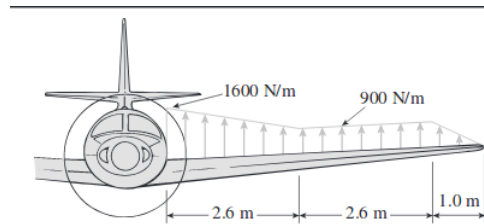


11. If the bending moment of a simply supported beam is as shown, the the load acting on the beam is

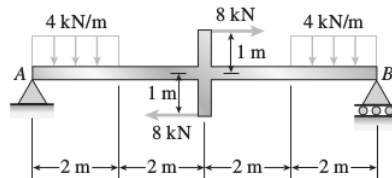


12. A simply supported beam is subjected to a couple somewhere in the span, it would produce.....
13. A cantilever with span L and Load F at the free end. The Bending Moment at the support will be
14. For the shear force to be uniform along the beam it should carry a anywhere in its span (*moment/couple*).
15. If a simply supported beam at its end is loaded by a central concentrated load then maximum moment is M . If the same weight is equally distributed over the beam then the maximum bending moment will be.....($M/2$)
16. The rate of change of Bending moment is equal to the at that section (*shear force*)
17. Point of contra-flexure is where changes sign
18. At the point of contra-flexure the shear force ins the shear force diagram will be
19. Bending moment variation due to concentrated loads will be(*Linear*)
20. The bending moment at a point on a beam is the algebraic sum of all the on either side of the point.(*moment*)

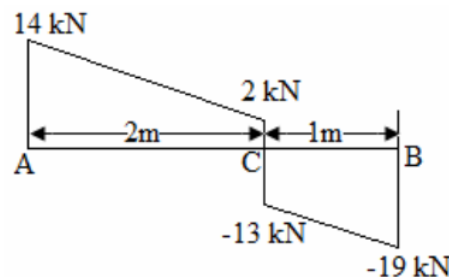
Main Questions



1. Under cruising conditions the distributed load acting on the wing of a small air plane has the idealized variation shown in the figure. Calculate the shear force V and bending moment M at the inboard end of the wing.
2. A simple beam AB is loaded by two segments of uniform load and two horizontal forces acting at the ends of a vertical arm (see figure). Draw the shear-force and bending-moment diagrams for this beam.

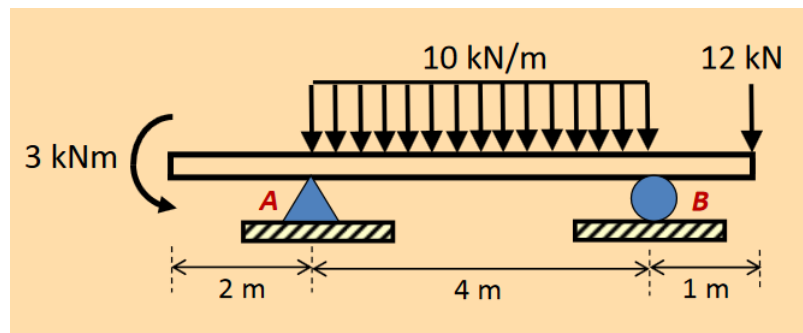


3. Derive the relationship between load, shear force and bending moment.
4. A Cantilever beam carrying gradually varying load from zero at fixed end and w /unit length at the free end. Draw the shear force and bending moment diagram.
5. Prove that the maximum bending moment is $wL^2/12$ for a simply supported beam of length l is subjected to a symmetrical uniformly varying load with zero intensity at the ends and intensity w (load per unit length) at the mid span.
6. Draw the loading conditions and the bending moment diagram for the given shear force diagram. Determine the boundary conditions of the beam.

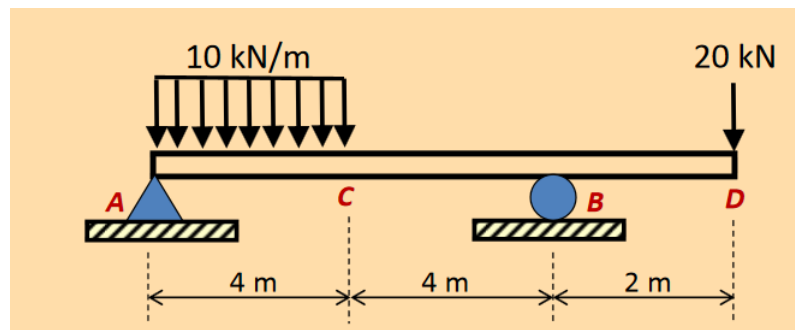


7. A Cantilever beam carrying gradually varying load from zero at fixed end and w /unit length at the free end. Draw the shear force and bending moment diagram. Also derive the relationship between load, shear force and bending moment. What can you conclude?

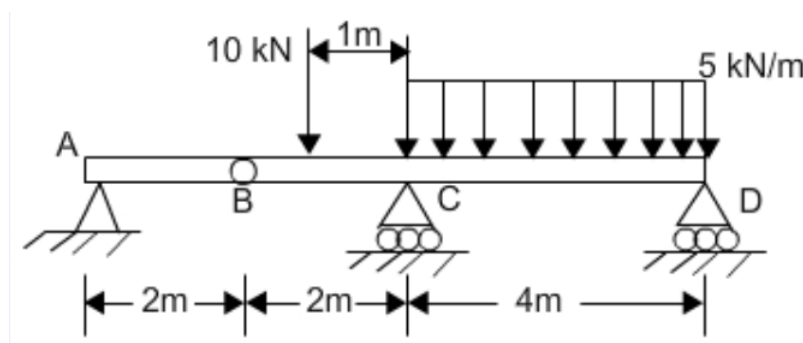
8. Prove that when a beam is subjected to a constant bending moment along its length, then the shear force will be zero at all sections along the beam.
9. A beam AB is hinged-supported at its ends and is loaded by couple $P \cdot c$ at the center of the beam. Determine the magnitude or shearing force at a section x . Also draw the shear force bending moment diagram for the same.
10. Prove that the maximum bending moment is $0.0645wl^2$ for a simply supported beam of length L subjected to a UDL Wx/L .
11. Calculate the shear force and bending moment for the beam(s) subjected to the loads as shown in the figure(s) then draw the shear force diagram (SFD) and bending moment diagram (BMD).



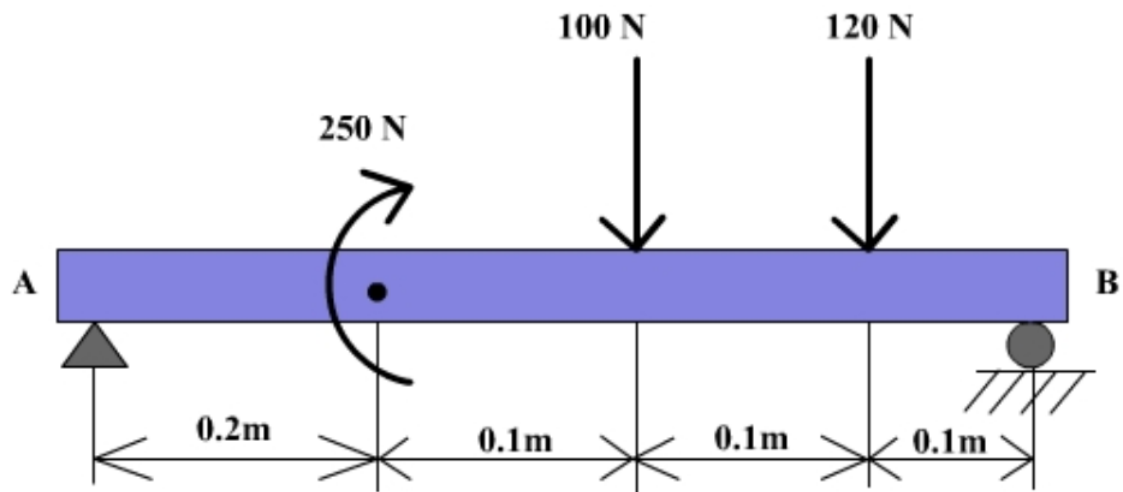
(a)



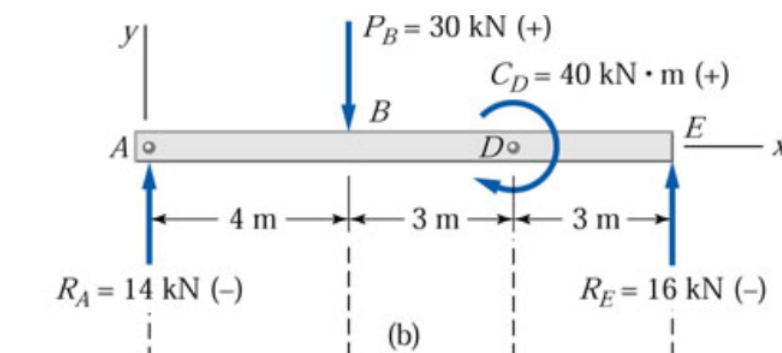
(b)



(c)

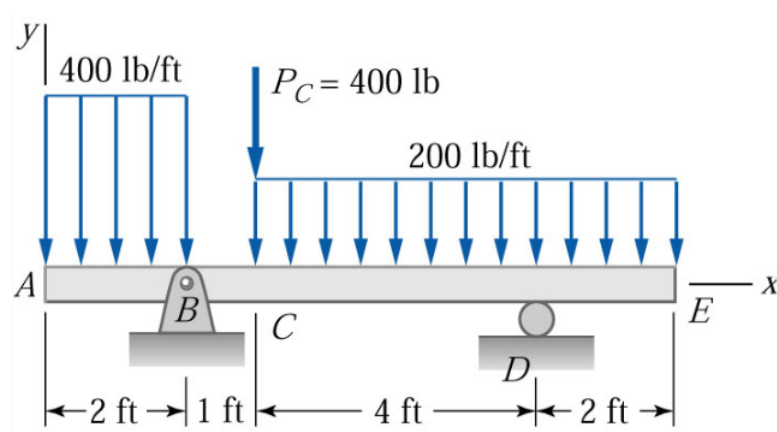


(d)



(b)

(e)



(f)

Chapter 3

Euler-Bernoulli beam theory and Deflection of beams

3.1 Question Bank UNIT - 3

Quiz Questions

1. Relationship between radius of curvature and curvature of the beam is
2. When the section modulus increases the bending stress
3. The axial force acting on the neutral axis is
4. Assumptions of the Euler Bernoulli beam are
5. A cantilever beam of rectangular cross-section is subjected to a load W at its free end. If the depth of the beam is doubled and the load is halved, the deflection of the free end as compared to original deflection will be....
6. In a cantilever beam, if the length is doubled while keeping the cross-section and the concentrated load acting at the free end the same, the deflection at the free end will increase by
7. Section modulus of a beam is always given by formula.....
8. Bending occurs due to the application of load
9. In cantilever beam, slope and deflection at free end is
10. *Deflection is maximum at a point where slope is zero*, this statement is true for a beam.
11. Moment of resistance is defined as
12. The area moment of inertia of a square of size 1 unit about its diagonal is
13. The term flexural rigidity can be written as
14. What are the conditions to be satisfied for determinate structures.....
15. In a cantilever beam if the length is doubled keeping the cross section and the concentrated load acting at the free end the same the deflection at the free end will increase by
16. In case of fixed beam subjected to gravity loads maximum hogging or negative bending moment develops at the(supports)

17. What are the units of flexural rigidity?
18. The maximum stresses occurs at top most fibre of a simply supported beam.
19. As compared to uniaxial tension or compression, the strain energy stored in bending is only (1/3)
20. When a rectangular beam is loaded longitudinally shear develops on fibres.

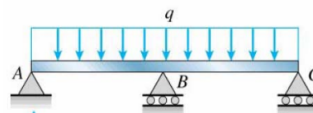
Main Questions

1. Derive the Differential Equation for the elastic curve of a beam.
2. If the beam cross-section is rectangular having a width of 75 mm, determine the required depth such that maximum bending stress induced in the beam does not exceed 40 MN/m^2
3. A simply supported beam of rectangular section 4 cm by 6 cm carries a mid-span concentrated load such that the 6 cm side lies parallel to line of action of loading; deflection under the load is . If the beam is now supported with the 4 cm side parallel to line of action of loading, the deflection under the load will be?
4. The deflection curve for a simply supported beam is given by the following equation:

$$W = -\frac{W_o \cdot x}{120LEI}(10L^3 - 10L^2 \cdot x + 5x^2 - x^3) \quad (3.1)$$

Describe the loading on the beam.

5. A beam with a rectangular section of 120 mm 60 mm, designed to be placed vertically is placed horizontally by mistake. If the maximum stress is to be limited, the reduction in load carrying capacity would be?
6. A continuous beam ABC supports a uniform load q . Determine the reactions. Assume the length between $AB = BC = L$.



7. Derive the complete Bending Equation and extend it to an expression for moment carrying capacity of a section.
8. Derive the expression for the deflection of a cantilever beam subjected to a point load at the free end using the double integration method.

9. A circular pipe of external diameter 70mm and thickness 8mm is used as a simply supported beam over an effective span of 2.5m. Find the maximum concentrated load that can be applied at the centre of the span if the permissible stress in the tube is 10N/mm^2
10. A simply supported beam, 2 in wide by 4 in high and 12 ft long is subjected to a concentrated load of 2000 lb at a point 3 ft from one of the supports. Determine the maximum fiber stress and the stress in a fiber located 0.5 in from the top of the beam at midspan.
11. Determine the maximum value of EI_y for the cantilever beam loaded as shown in Fig. Take the origin at the wall.

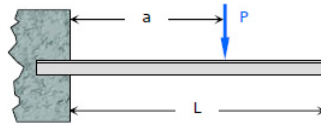


Figure P-607

12. Derive the expressions for beams subjected to combined loads inclined loads and eccentric loads.

Chapter 4

Torsion

4.1 Question Bank UNIT - 4

Quiz Questions

1. The diameter of a shaft is increased from 30 mm to 60 mm, all other conditions remaining unchanged. How many times is its torque carrying capacity increased?
2. Angle of twist of a shaft of diameter 'd' is inversely proportional to(in terms of 'd')
3. Shear strain in shaft subjected to torsion is inversely proportional to
4. The ratio of Min shear stress to Max shear stress for a thin walled closed structure is
5. A solid shaft transmits a torque T. The allowable shearing stress is τ . What is the diameter of the shaft?
6. A hollow shaft of outer dia 40mm and inner dia of 20mm is replaced by a solid shaft to transmit the same torque at the same maximum stress. What should be the diameter of the solid shaft?
7. A circular shaft is subjected to the combined action of bending, twisting and direct axial loading. The maximum bending stress σ , maximum shearing force 3σ and a uniform axial stress σ (compressive) are produced. The maximum compressive normal stress produced in the shaft will be.
8. One-half length of 50 mm diameter steel rod is solid while the remaining half is hollow having a bore of 25 mm. The rod is subjected to equal and opposite torque at its ends. If the maximum shear stress in solid portion is τ or, the maximum shear stress in the hollow portion is:
9. Torsional rigidity is defined as
10. The ratio of torque carrying capacity of a solid shaft to that of a hollow shaft is given by.....

Main Questions

1. A solid circular rod AB of diameter D and length L is fixed at both ends. A torque T is applied at a section X such that $AX = L/4$ and $BX = 3L/4$. What is the maximum shear stress developed
2. In a torsion test, the specimen is a hollow shaft with 50 mm external and 30 mm internal diameter. An applied torque of 1.6 kN-m is found to produce an angular twist of 0.4° measured on a length of 0.2 m of the shaft. The Youngs modulus of elasticity obtained from a tensile test has been found to be 200 GPa. Find the values of (i) Modulus of rigidity. (ii) Poissons ratio.
3. A hollow steel rod 200 mm long is to be used as torsional spring. The ratio of inside to outside diameter is 1 : 2. The required stiffness of this spring is 100 N.m /degree. Determine the outside diameter of the rod. Value of G is 8×10^4 N/mm²
4. A hollow steel rod 200 mm long is to be used as torsional spring. The ratio of inside to outside diameter is 1 : 2. The required stiffness of this spring is 100 N.m /degree. Determine the outside diameter of the rod. Value of G is 8×10^4 N/mm²
5. Calculate the maximum intensity of shear stress induced and the and the angle of twist in degrees in a solid shaft of 100 mm diameter, 10m long, transmitting a power of 112.5KW at 150 rpm. Take $G=82$ KN/mm²
6. A solid shaft is to transmit 300 kW at 120 rpm. If the shear stress is not to exceed 100 MPa, Find the diameter of the shaft, What percent saving in weight would be obtained if this shaft were replaced by a hollow one whose internal diameter equals 0.6 of the external diameter, the length, material and maximum allowable shear stress being the same?
7. Derive the Complete Torsion equation. Determine the torsional stiffness of a hollow shaft of length L and having outside diameter equal to 1.5 times inside diameter d . The shear modulus of the material is G .
8. A hollow shaft of diameter ratio 3/8 required to transmit 600 kW at 110 rpm, the maximum torque being 20% greater than the mean. The shear stress is not to exceed 63 MPa and the twist in a length of 3 m not to exceed 1.4 degrees. Determine the diameter of the shaft. Assume modulus of rigidity for the shaft material as 84 GN/m².
9. A 5 cm diameter solid shaft is welded to a flat plate by 1 cm filled weld. What will be the maximum torque that the welded joint can sustain if the permissible shear stress in the weld material is not to exceed 8 kN/cm² ? Deduce the expression for the shear stress at the throat from the basic theory.
10. A hollow shaft and a solid shaft construction of the same material have the same length and the same outside radius. The inside radius of the hollow shaft is 0.6 times of the outside radius. Both the shafts are subjected to the same torque. (i) What is the ratio of maximum shear stress in the hollow shaft to that of solid shaft? (ii) What is the ratio of angle of twist in the hollow shaft to that of solid shaft?

Chapter 5

Thin walled cylinders and Failure theories

5.1 Question Bank UNIT - 5

Quiz Questions

1. Which theory of failure will you use for aluminium components under steady loading?
2. Design of shafts made of brittle materials is based on theory.
3. Hoop stress for a thin cylinder is given by.....
4. The ratio of circumferential stress to longitudinal stress in a thin cylinder subjected to internal hydrostatic pressure is:.....
5. Who postulated the maximum distortion energy theory?
6. A transmission shaft subjected to bending loads must be designed, on the basis of theory.
7. From design point of view, spherical pressure vessels are preferred over cylindrical pressure vessels because.....
8. For the same internal diameter, wall thickness, material and internal pressure, the ratio of maximum stress, induced in a thin cylindrical and in a thin spherical pressure vessel will be:.....
9. A thin walled spherical shell is subjected to an internal pressure. If the radius of the shell is increased by 1% and the thickness is reduced by 1%, with the internal pressure remaining the same, the percentage change in the circumferential (hoop) stress is
10. A thin cylinder of 100mm internal diameter and 5mm thickness is subjected to an internal pressure of 10 MPa and a torque of 2000 Nm. Calculate the magnitudes of the principal stresses.

Main Questions

1. Derive the expression for Circumferential Stress and Longitudinal stress in a thin cylinder.
2. The stress state at a point in a body is plane with $\sigma_1 = 60 \text{ N/mm}^2$ and $\sigma_2 = 36 \text{ N/mm}^2$. If the allowable stress for the material in simple tension or compression is 100 N/mm^2 calculate the value of factor of safety with each of the following criteria for failure (i) Max Stress Criteria (ii) Max Shear Stress Criteria (iii) Max strain criteria (iv) Max Distortion energy criteria
3. A cylindrical shell is 3m long and having 1m internal diameter and 15mm thickness. Calculate the maximum intensity of shear stress induced and also the changes in the dimensions of the shell if it is subjected to an internal fluid pressure of 1.5 N/mm^2
4. Explain with suitable equations Coulomb Failure Theory. For which materials this theory is best suited and why? Discuss.
5. A cylindrical shell has the following dimensions: Length = 3 m Inside diameter = 1 m Thickness of metal = 10 mm Internal pressure = 1.5 MPa Calculate the change in dimensions of the shell and the maximum intensity of shear stress induced. Take $E = 200 \text{ GPa}$ and Poissons ratio = 0.3
6. State the Von- Mises's theory. Write the equation of criteria for failure.
7. Calculate the maximum principal strain in a thin cylindrical tank, having a radius of 25 cm and wall thickness of 5 mm when subjected to an internal pressure of 1MPa, (take Young's modulus as 200 GPa and Poisson's ratio as 0.2)
8. A bolt is subjected to an axial pull of 12kN together with a transverse shear force of 6kN. Determine the diameter of the bolt by using (a) Maximum Principal Stress Theory, (b) Maximum strain theory, (c) Maximum shear stress theory. Take Elastic limit in tension to be 300 N/mm^2 Factor Of Safety=3 and poissons ratio=0.3
9. Describe the following theories of failure. 1. Max Shear Stress Theory 2. Maximum Principal Strain theory.
10. A cylinder thin shell of 800mm diameter and 3m long has a shell thickness of 10mm. If the shell is subjected to an internal pressure of 2.5 N/mm^2 , determine the change in diameter, length and volume.